

Mohammad Sheharyar

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ABOUT MYSELF

Motivated Software Engineer doing Data Analysis, Machine Learning Deployment, and Back-End Development. Proficient in Python and Node.js for server-side logic and database management. Strong understanding of Google Cloud Platform (GCP). Experienced in SQL and Pandas for data wrangling and analysis in BigQuery, Jupyter, Kaggle, Collaboratory Notebooks, or Microsoft Access. Possesses knowledge of R's Tidyverse package. Utilizes Tableau to create industry-standard visualizations and dashboards for communicating insights on business tasks through Microsoft PowerPoint presentations. Addresses Business Intelligence processes with spreadsheets (Microsoft Excel and Google Sheets). Employs Power Query and VBA Macros in Excel & Power BI for Data ETL automation.

WORK EXPERIENCE

 **IDEVNERDS** – LAHORE, PAKISTAN
Business or Sector Information and communication | **Website** <https://idevnerds.com/>

SOFTWARE ENGINEER – 01/2025 – 04/2025

- Back-End Web Dev Intern
 - Built complete RESTful APIs with Express.js for an e-commerce platform
 - Optimized APIs using async processing (non-blocking) and leveraging connection pooling with ORM/ODM
 - Wrote Middleware to implement JSON Web Token Authentication in cookies for separate client and admin routes
 - Wrote Middleware to validate data using the Joi library before insertion into databases
 - Encrypted data with the bcrypt library
 - Created Dockerfiles for deploying containerized databases
 - Used Docker CLI for managing images and containers
 - Used Sequelize ORM and CLI to interface with MySQL and PostgreSQL databases
 - Used mysql and psql CLIs, MySQL Workbench & PGAdmin
 - Used Mongoose ODM to interface with MongoDB databases
 - Used the mongosh CLI & MongoDB Compass
 - Tested APIs with curl and Postman (VS Code extension)
 - Managed project packages using Node Package Manager (npm)

 **MACHKNOWSOFT** – ABBOTTABAD, PAKISTAN
Business or Sector Information and communication | **Website** <https://www.machknowsoft.net/>

SOFTWARE ENGINEER – 05/2024 – 11/2024

- Software Development Intern
 - Created Python application software
 - Created GUIs using the Python tkinter library
 - Created websites/web apps using Flask micro-framework
 - Wrote test cases and unit tests for apps
 - Created scripts for Linux Bash using Python os, shutil, and psutil modules
 - Managed project packages using Python Package Manager (pip)

PROJECTS

Social Media Addiction Exploratory Data Analysis

- Used Python, Pandas & Matplotlib to check, clean, analyze, and visualize an open, high-quality dataset sourced on Kaggle
- Created hypotheses to test
- Produced statistics for the dataset
- Ensured the statistical significance of the analysis
- Delivered insights on how Social Media addiction impacts young adults globally using Matplotlib plots

Link <https://www.kaggle.com/code/mohammadsheharyar/social-media-addiction-eda>

FDA Food Events Exploratory Data Analysis

- Used SQL with the Google BigQuery API to check, clean, analyze, and visualize an open, high-quality dataset sourced from BigQuery public datasets
- Utilized Window Functions, aggregations, subqueries, JOINS, and temporary tables
- Analyzed the number of serious health events caused by the food industries

- Analyzed which food categories severely affected infants under the age of 3
- Plotted the data of affected infants
- Analyzed the number of fatal outcomes over 25 years

Link <https://www.kaggle.com/code/mohammadsheharyar/fda-food-events-eda>

Gen AI course Capstone Project

- Utilized Google Gen AI Python SDK to develop a solution for AI-generated Subtitles for the Deaf and Hard of Hearing (SDH) in a structured JSON format with few-shot prompting
- Interpreted American Sign Language (ASL) using the grounding feature to ensure accuracy instead of a Retrieval Augmented Generation (RAG) solution

Link <https://www.kaggle.com/code/mohammadsheharyar/sdh-generation-with-google-gen-ai>

Data Analysis Project

- Used the Pandas and NumPy libraries to generate and then simulate the analysis of Social Media post categories
- Cleaned data with dataframe methods
- Used dataframe methods and Matplotlib plots to visualize and analyze data

Link <https://github.com/M0S111/Coursera-DataAnalysis-Project/blob/main/SocialMediaDataAnalysis.ipynb>

World Bank/UNWTO Tourism data (1995-2020)

- Sourced open data from the World Bank data bank for national receipts from and expenditure on tourism
- Retained original data in separate sheets
- Sorted, filtered, and applied conditional formatting on the data to clean it of missing values
- Created pivot tables to consolidate relevant data for 5 years (2016-2020)
- Created basic bar graph visualization for receipts and expenditures

Link https://docs.google.com/spreadsheets/d/1bVY-9Gn5vquglRcGC_jtQwGvK0KWA3NCS_TJ89kQ7-g/edit?usp=sharing

Visualization of the World Bank/UNWTO Tourism data (2020)

- Cleaned and formatted the World Bank/UNWTO open dataset in the spreadsheets mentioned above
- Visualized the receipts in millions of dollars for international tourism during 2020 as a heatmap
- Created an accompanying bar graph of the same

Link <https://public.tableau.com/app/profile/mohammad.sheharyar/viz/WorldBankUNWTOTourism2016-2020/Sheet2>

Visualization of Movies Performance

- Created a dashboard of the profitability of movie genres during 2 years (2014-2016)
- Visualized genre profitability over time as a line graph
- Showed the revenue earned by famous actors as a circle packing diagram
- Charted the number of movies actors acted in per genre as a stacked bar graph

Link <https://public.tableau.com/app/profile/mohammad.sheharyar/viz/MoviesFinancialPerformance2014-2016/Dashboard1>

Personal Website

- Used Synchronized Multimedia Integration Language (SMIL) with HTML SVG elements to create an unorthodox navigation menu
- Used jQuery animate() and css() methods with CSS keyframe animations to create a neon glow effect
- Created a simple loop-around carousel to display certificates
- Called an external jokes API using JavaScript Fetch API to serve programming jokes from JSON payloads
- Shortened page loading time with lazy loading of images

Link <https://m0s111.github.io/>

Recommendation System (University Capstone)

- Created a hybrid recommendation system for Western movies and deployed it as a single-page application (SPA)
- Implemented the system as a Machine Learning model using the k-Nearest Neighbors algorithm using the scikit-learn library
- Stored the model as a serialized Pickle file
- Ran the model as a Flask app on a WSGI server

Links https://github.com/M0S111/Rec-Sys/blob/main/Rec_Sys.ipynb | <http://msheharyar111.pythonanywhere.com/>

● **EDUCATION AND TRAINING**

09/2018 – 09/2023 Lahore, Pakistan
BACHELOR OF SCIENCE Virtual University of Pakistan

Website <https://vu.edu.pk> | **Field of study** Software and applications development and analysis | **Final grade** 3.65 | **Level in EQF** EQF level 6

● **SKILLS**

Data Analysis

Jupyter Notebooks | SQL | Pandas | Tableau | Spreadsheets | Relational databases | Google BigQuery | Google Cloud Dataflow | Microsoft Excel | Microsoft Access | Microsoft PowerPoint | Microsoft Power BI | Microsoft Power Query | Artificial Intelligence (AI)

Web Development

JavaScript | ECMAScript | Node.js | Express.js | Python | Flask | Git | Github | HTML | CSS | Linux | JSON | REST | RESTful APIs | Postman | MySQL | PostgreSQL | NoSQL | MongoDB | Docker | Object-Oriented Programming | Unit Testing | Design Patterns | Google Cloud Platform | Agile (Scrum)

● **CERTIFICATES**

Google Data Analytics

Link <https://coursera.org/verify/professional-cert/3ECBW1QJD9N6>

Google IT Automation with Python

Link <https://coursera.org/verify/professional-cert/K5LZWFKPHB9F>

Google Project Management

Link <https://coursera.org/verify/professional-cert/L5XJ3J2MTCPL>

UC San Diego ALGS200x: Algorithmic Design and Techniques

Link <https://courses.edx.org/certificates/778698a318064eb2b48c950684885c9c>

UC San Diego ALGS201x: Data Structures Fundamentals

Link <https://courses.edx.org/certificates/df74a80d90e74d1a80c4b839737748e6>

● **LANGUAGE SKILLS**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C1
JAPANESE	B1	B1	B1	B1	A2

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user